

Syntax

- The **signature** $\mathcal{L}$ now consists of:
 - set of sorts, denoted by α, β
 - constants, denoted by a, b, c , each having a sort $c : \alpha$
 - function symbols, denoted by f, g , each having a type $\alpha_1 \times \dots \times \alpha_n \mapsto \alpha$
 - predicate symbols, denoted by p, q , each having a type $\alpha_1 \times \dots \times \alpha_n$
- **Quantifiers**:
 - Given a formula G , a variable x , then $\mathcal{Q}x : \alpha A$ is a quantified formula with $\mathcal{Q} \in \{\forall, \exists\}$
 - A formula is called quantifier-free (or ground) if it contains no occurrences of quantifiers

Semantics

- The **interpretation** I maps:
 - Each sort α to a non-empty set $I(\alpha)$, called domain of the sort
 - Each constant $c : \alpha$ to $c' \in I(\alpha)$
 - Each variable $x : \alpha$ to $x' \in I(\alpha)$
 - Each function symbol $f : \alpha_1 \times \dots \times \alpha_n \mapsto \alpha$ to a function $f' : I(\alpha_1) \times \dots \times I(\alpha_n) \mapsto I(\alpha)$
 - Each predicate symbol $p : \alpha_1 \times \dots \times \alpha_n$ to a relation $p' \subseteq I(\alpha_1) \times \dots \times I(\alpha_n)$
- Given I and a term t of a sort α , define the element $t' \in I(\alpha)$ as follows:
 - For constants and variables $t' = I(t)$
 - $f(t_1, \dots, t_n)' = f'(t'_1, \dots, t'_n)$
- Given I and a formula G , define the boolean value as follows:
 - For an atomic formula $p(t_1, \dots, t_n)$, define $p(t_1, \dots, t_n)' = 1 = (t'_1, \dots, t'_n) \in p'$
 - $G = (Q_1 \circ Q_2)' = 1$ if $Q'_1 \circ Q'_2$ is satisfied
- **Quantifiers**
 - Given a set sequence of variables $\vec{x}$, two interpretations of the same signature are $\vec{x}$ -variant if they coincide on all symbols and variables not $\vec{x}$.
 - $(\forall xA)' = 1$ if for all $\vec{x}$ -variants I' of I we have $(A)' = 1$
 - $(\exists xA)' = 1$ if for some $\vec{x}$ -variant I' of I we have $(A)' = 1$

Axiomatic Interpretations

Restrict interpretations to those which satisfy the following set of formulas (so called axioms):

- reflexivity : $x = x$
- symmetry : $x = y \mapsto y = x$
- transitivity : $x = y \wedge y = z \mapsto x = z$
- congruence axioms for each function symbol f :
$$\left(\bigwedge_{i=1}^n x_i = y_i\right) \mapsto f(x_1, \dots, x_n) = f(y_1, \dots, y_n)$$
- congruence axioms for each predicate symbol p :
$$\left(\bigwedge_{i=1}^n x_i = y_i \wedge p(x_1, \dots, x_n)\right) \mapsto p(y_1, \dots, y_n)$$

Interpretation Classes

Restrict interpolations to those which satisfy certain properties

- **Example** :
 - Given the formula $f(x) = f(x + 1) \wedge f(a) \neq f(b)$
 - 1.We consider the class of interpretations I where I interprets $+$ and 1 in the standard way over the integers. The formula is satisfiable in:
$$I(a) = 1, I(b) = 2, I(x) = 3$$
$$f(x) = \begin{cases} 3 & \text{if } x = 4 \\ x & \text{else} \end{cases}$$
 - 2.We consider the class of interpretations of example 1. but where $f(x) = f(x + 1)$ is valid in I . The formula is not satisfiable as $f(x) = f(x + 1)$ would impose that f maps all variables to the same value, in which case $f(a) \neq f(b)$ can not hold.
- Satisfiability and Interpretation - Examples**
- Given the formula $x = y + 1$ over integers, where x, y are variables, $+$, 1 are interpreted in the standard way over integers and the interpretation $I(x) = I(y) = 0$. It then holds that:
 - $I \not\models x = y + 1$
 - $x = y + 1$ is satisfiable in I under the $\vec{x}$ -variant: $I'(x) = 1, I(y) = 0$
 - $x = y + 1$ is not valid in I , since $I \not\models x = y + 1$
 - Given the formula $z \neq a \mapsto z = b$, where $a : \alpha, b : \alpha$ are constants and $z : \alpha$ is a variable. This formula is valid under the class of interpretations I for which it holds that:
 - $D^I_\alpha = \{0, 1\}, I(a) = 0, I(b) = 1$
 - Given the formula $y = a \mapsto y = b$, where a, b are constants and y is a variable. This formula is valid under the class of interpretations I for which it holds that :
 - $I(a) = I(b)$

Example :

Given based on table to the right:

- $I \models F_1 \wedge F_2 \wedge F_3 \wedge F_4$
- $x, y, c_1, c_2, c_3, c_4 : \alpha ; f : \alpha \mapsto \alpha$

Let:

- $I(\alpha) := \mathbb{Z}, I(\alpha), I(y), I(c_1), I(c_2), I(c_3), I(c_4) \in \alpha = \mathbb{Z}$
- $I(f) : \mathbb{Z} \mapsto \mathbb{Z}, I(\alpha) : \mathbb{Z} \mapsto \mathbb{Z}$ (which means, that array a is interpreted as a function)

Set:

- $I(c_1) = 4, I(c_4) = 3, I(c_2) = I(c_3) = 0, I(y) = I(x) + 2 = 2$
- $I(\alpha)(i) := 0$ forall $i \in \mathbb{Z}$
- $I(f)(v) = \begin{cases} 1 & v = 4 \\ 0 & \text{else} \end{cases}$

Theory

A theory is defined as

- a signature Σ_T
- a set of formulas A_T over Σ_T , called axioms of T

Unless specified otherwise, theories are quantifier-free

- An interpretation in which A_T are valid is called a T -interpretation
- If a formula G is valid in every T -interpretation, then G is T -valid (denoted by $T \models G$)

Theory of equality $\overline{T}_E$:

- $\Sigma_E = \{a, b, \dots, f, g, \dots, p, =, \dots\}$
- $A_E = \left\{ \begin{aligned} &x = x, x = y \mapsto y = x, x = y \wedge y = z \mapsto x = z, \\ &\bigwedge_{i=1}^n y_i = x_i \mapsto f(x_1, \dots, x_n) = f(y_1, \dots, y_n), \\ &\bigwedge_{i=1}^n y_i = x_i \wedge p(x_1, \dots, x_n) \mapsto p(y_1, \dots, y_n) \end{aligned} \right\}$

A binary relation R over a set S is called a congruence relation, if:

- R is reflexive, transitive, symmetric and
- $\bigwedge_{i=1}^n y_i R x_i \mapsto f(x_1, \dots, x_n) R f(y_1, \dots, y_n)$

The congruence class of $t \in S$ under R is $[t]_R = \{t' \in S \mid t R t'\}$,

for which holds:

- $\bigcup_{[t]_R} [t]_R = S$
- $[t_1]_R \neq [t_2]_R \mapsto [t_1]_R \cap [t_2]_R = \emptyset$

The congruence closure R^c of R is the relation for which holds:

- $R \subseteq R^c$
- $\forall R' : R \subseteq R' \mapsto (R^c = R' \vee R^c \subseteq R')$

Congruence Closure Algorithm for $\overline{T}_E$

```
procedure CongruenceClosure(F)
input:  $\overline{F}$  is  $s_1 = t_1 \wedge \dots \wedge s_n = t_n \wedge s_{n+1} \neq t_{n+1} \wedge \dots \wedge s_m \neq t_m$ 
output: satisfiable or unsatisfiable
parameters: function subterm.set
begin
 $S_F := \text{subterm.set}(F)$ 
 $R := \{sRs \mid s \in S_F\}$  and  $[s]_R := \{s\}$ , defining the partition  $\{[s]_R \mid s \in S_F\}$ 
for every  $s_i = t_i$  in  $\overline{F}$ , merge  $[s]_R$  and  $[t]_R$  by
    forming the union  $[s]_R \cup [t]_R$ 
    propagate the new congruences that arise in this union (function congruence)
if  $s_i R t_i$  for any  $j \in \{n+1, \dots, m\}$  then return unsatisfiable
else return satisfiable end
```

Example

- Given the formula $F : f(a, b) = a \wedge f(f(a, b), b) \neq a$
- We proceed as follows:
- 1.Subterm Set $S_F = \{a, b, f(a, b), f(f(a, b), b)\}$
 - 2.Initial partition $P = \{\{a\}, \{b\}, \{f(a, b)\}, \{f(f(a, b), b)\}\}$
$$P = \{\{a, f(a, b)\}, \{b\}, \{f(f(a, b), b)\}\}$$
 - 3.Given $f(a, b)Ra$, we merge the partition :
$$P = \{\{a, f(a, b)\}, \{b\}, \{f(f(a, b), b)\}\}$$
 - 4. Given $f(a, b)Ra \wedge bRb \mapsto f(f(a, b), b)Rf(a, b) :$
$$P = \{\{a, f(a, b), f(f(a, b), b)\}, \{b\}\}$$
 - 5.Due to the clause $f(f(a, b), b) \neq a$, formula F is $\overline{T}_E$ -unsatisfiable, since $f(f(a, b), b) \in [a]_R \iff f(f(a, b), b) R a \in R$, but the formula states, that $f(f(a, b), b) \neq a$

DPLL(T) - For Non-Unit Clauses in $\overline{T}_E$

Given a formula $G := (a = b \vee a = c) \wedge \dots$ then G contains a non-unit clause and can thus not be proven with congruence closure. Instead, we DPLL $_T$:

```
procedure DPLL $_T$ (F)
input: A CNF formula  $F$  in the decidable theory  $T$ 
output: satisfiable or unsatisfiable
parameters: functions propositional.abstraction, decision.procedure $_T$ 
begin
 $S := \text{propositional.abstraction}(F)$ 
while true do
if  $DPLL(S) = \text{unsatisfiable}$  then return unsatisfiable
if  $DPLL(S) = \text{satisfiable}$  then
 $L := \{L_1, \dots, L_n\}$ , s.t. for the model  $I$  of  $L$  we have  $I \models S$ 
if decision.procedure $_T$ ( $L$ ) = satisfiable then return satisfiable
if decision.procedure $_T$ ( $L$ ) = unsatisfiable then
 $S := S \cup \{\neg L_1, \vee \dots \vee \neg L_n\}$ 
end do
end
```

- Example

Given the formula $F :$
 $(a = b \vee a = c) \wedge f(a) = a \wedge f(f(b)) \neq a \wedge f(c) \neq a$

1.Add propositional symbol for every theory atom:

- $p_1 : a = b, p_2 : a = c, p_3 : f(a) = a,$
- $p_4 : f(f(b)) \neq a, p_5 : f(c) = a$
- $F_P = (p_1 \vee p_2) \wedge p_3 \wedge \neg p_4 \wedge \neg p_5$

2.If F_P is unsat, then so is F

3.If F_P is sat, then we have literals $L_1, \dots, L_n$ of I for which holds: $I \models F_P$
e.g. $p_1, \overline{p}_2, p_3, \overline{p}_4, \overline{p}_5$

4.Use clauses corresponding to literals and check with congruence closure

$F' = a = b \wedge a \neq c \wedge f(a)$
since $\overline{p}_3$

5.If F' is T -sat, terminate, else define $F_P = F_P \wedge (\neg L_1 \vee \dots \vee \neg L_2)$ and go to 2.

Nelson Oppen method

Given $T_1, \dots, T_n$ decidable, quantifier-free theories with disjoint signature and which are stably infinite the Nelson Oppen Method for **non-convex** theories finds disjunctions of new equalities $\bigvee u_k = v_k$ for shared

u_k, v_k . Split disjunction and exchange equalities along multiple branches to other theories.

Linear Arithmetic (F_1)	Uninterpreted Functions (F_1) $f(c_1) \neq f(c_1)$	Linear Arithmetic (F_1)	Uninterpreted Functions (F_2) $f(c_1) \neq f(c_1)$
$y = x + 2$ $c_1 = y - x + 2$ $c_2 = y - 2$ $c_3 = 3$		$y = x + 2$ $c_1 = y - x + 2$ $c_2 = y - 2$ $c_3 = 3$	
		$c_4 = 3$ $c_1 = 4$	
		Arrays (F_3) $c_4 = \text{read}(\text{write}(a, x, c_1), c_2)$	Shared Equalities (F_4) $c_2 = x$ $c_4 = c_3$

Theory of arrays $T_A :$

- $\Sigma_A = \{\text{read}, \text{write}\}$
where *read* is a binary function and *write* is ternary function
- $A = A_E \cup \left\{ \begin{aligned} &x = y \mapsto \text{read}(\text{write}(A, x, v), y) = v, && \text{r-o-w}_1 \\ &x \neq y \mapsto \text{read}(\text{write}(A, x, v), y) = \text{read}(A, y) && \text{r-o-w}_2 \end{aligned} \right\}$

- $\text{read}(A, x)$ denotes the value of array A at pos. x
- $\text{write}(A, x, v)$ denotes the modified array A with value v at pos. x

Given a function F with *write*-terms $\text{read}(\text{write}(A, x, v), y)$, $F[F']$ denotes the formula where the *write*-term has been replaced with F'

Congruence Closure Algorithm for T_A

T_A -satisfiability of F is reduced to $\overline{T}_E$ -satisfiability:

- If F contains *write*-terms $\text{read}(\text{write}(A, x, v), y)$, use r-o-w axioms:

- 1. $F_1 : x = y \wedge F[v]$
- 2. $F_2 : x \neq y \wedge F[\text{read}(A, y)]$
- If both are unsat, then F is unsat

- If F doesn't contains *write*-terms:
 - Use fresh function symbol f_A for array A
 - Replace all $\text{read}(A, x)$ with $f_A(x)$

Return $\overline{T}_E$ satisfiability of modified formula

Combination of Theories

Given:

- Given a set of theories $T_1, \dots, T_n$ in disjoint $\Sigma_1, \dots, \Sigma_n$ and its axioms $\bigcup_{i=1}^n A_{T_i}$

- Quantifier-free conjunction F of literals in $\bigcup_{i=1}^n \Sigma_i$

Checking if F is satisfiable in a $T :$

- Transform $F : \bigwedge_{i=1}^n F_i$ with F_i being a T_i formula, where $F_1, \dots, F_n$ are linked by variables

- Example:

Given $T_A, \overline{T}_E$ and $F : (y = x + 2) \wedge (f(\text{read}(\text{write}(a, x, 3), y - 2)) \neq f(y - x + 1))$, we split:
 $F : (y = x + 2) \wedge f(c_1) \neq f(c_1)$

Linear Arithmetic $y = x + 2$ $c_1 = y - x + 1$ $c_2 = y - 2$ $c_3 = 3$	Uninterpreted Functions $f(c_1) \neq f(c_1)$
Arrays $c_4 = \text{read}(\text{write}(a, x, c_2), c_2)$	Shared Equalities $c_2 = x$ $c_4 = c_3$ $c_1 = c_2$

$\neg c_4 = \text{read}(\text{write}(a, x, c_2), c_2) \wedge c_2 = c_x$
r-o-w $_1$ axiom
 $\implies c_4 = c_3$

$f(c_4) \neq f(c_1) \implies c_4 \neq c_1$
 $c_4 = c_3$
 $c_3 = c_1$
 $\implies$ Contradiction! The formula is unsat

Split Reasoning Algorithm

```
procedure SeparatingReasoning(F)
input: formula  $F \in T_1 \cup \dots \cup T_n$ 
output: formulas  $F_1 \in T_1, \dots, F_n \in T_n$  s.t.
 $F = F_1 \wedge \dots \wedge F_n$  are equisatisfiable
assumptions: theory signatures  $\Sigma_1, \dots, \Sigma_n$  are disjoint
parameters: function head(t) returning the root symbol of a term  $t$ 
begin
repeat as long as possible
if  $t \in \Sigma_i$  and  $\text{head}(t) \in \Sigma_j$  with  $i \neq j$ :
    rewrite  $F[t(t_1, \dots, t_n)]$  into  $F[f(t_1, \dots, c, \dots, t_n)] \wedge c = t$ ,
    where  $c$  is a fresh new variable
if  $\rho \in \Sigma_i$  and  $\text{head}(t) \in \Sigma_j$  with  $i \neq j$ :
    rewrite  $F[\rho(t_1, \dots, t_n)]$  into  $F[\rho(t_1, \dots, c, \dots, t_n)] \wedge c = t$ ,
    where  $c$  is a fresh new variable
if  $\text{head}(s) \in \Sigma_i$  and  $\text{head}(t) \in \Sigma_j$  with  $i \neq j$ :
    rewrite  $F[s = t]$  into  $F[s = c] \wedge c = t$ ,
    where  $c$  is a fresh new variable
end repeat
return modified  $F$  as  $F_1 \wedge \dots \wedge F_n$  with each  $F_i \in T_i$ 
```

Not all theories can be easily combined this way

Convex Theory

A formula $F \in T$ as a conjunction of T -literals is convex if:

- $T \models \left(F \mapsto \bigvee_{j=1}^k (u_j = v_j) \right) \implies T \models F \mapsto u_j = v_j$
for some $j \in \{1, \dots, k\}$

- Example

T_A e.g. is not convex, $\overline{T}_E$ e.g. is convex

(Proof of T_A not being convex is on the next page)

Stably infinite theories

A theory T_i is called stably infinite:

- For every T satisfiable formula $F \in T$, there exists a T -interpretation I :
 - $I \models F$
 - I has a domain of infinite cardinality

$\overline{T}_E, T_A, \overline{T}_E$ are stably infinite

Nelson Oppen method

Given $T_1, \dots, T_n$ decidable, quantifier-free theories with disjoint signature and which are stably infinite the Nelson Oppen Method for **convex** theories finds new equalities $u = v$ for shared u, v . Pass equality to other theories.

```
procedure NelsonOppenConvex(F)
input: conjunctive formula  $F \in T_1 \cup \dots \cup T_n$ 
output: satisfiable or unsatisfiable
assumptions: Nelson-Oppen restrictions over  $T_i$  hold
assumptions:  $F_i$  are convex
begin
1.  $F_1 \wedge \dots \wedge F_n := \text{SeparatingReasoning}(F)$ 
2. Run  $T_i$ -solver on  $F_i$ . If a solver returns unsatisfiable, return unsatisfiable
3. If there are shared variables  $u, v$  s.t.  $F_i \mapsto u = v$  then
 $F_i := F_i \wedge u = v$ 
go to step 2
return satisfiable
end

procedure NelsonOppenNonConvex(F)
input: conjunctive formula  $F \in T_1 \cup \dots \cup T_n$ 
output: satisfiable or unsatisfiable
assumptions: Nelson-Oppen restrictions over  $T_i$  hold
assumptions:  $F_i$  are non-convex
begin
1.  $F_1 \wedge \dots \wedge F_n := \text{SeparatingReasoning}(F)$ 
2. Run  $T_i$ -solver on  $F_i$ . If a solver returns unsatisfiable, return unsatisfiable
3. If there are shared variables  $u, v$  s.t.  $F_i \mapsto u = v$  then
 $F_i := F_i \wedge u = v$ 
go to step 2
4. If  $\overline{F}_i \mapsto \bigvee_k u_k = v_k$ , for shared  $u_k, v_k$ , then
if NelsonOppenNonConvex( $F_1 \wedge \dots \wedge F_n \wedge u_k = v_k$ ) is satisfiable, for any  $k$ 
then return satisfiable
else return unsatisfiable
return satisfiable
end
```